

Year 8 Science – Booklet 17: Physics

Motion & Pressure — ANSWERS

Graphs of Motion – Quick Test (p.89)

1. On a distance-time graph, what do these show?

- a). A horizontal line: the object is not moving (stationary).
- b). A steeply sloping straight line: the object is moving at a high, constant speed.
- c). A straight line sloping just a little: the object is moving at a low, constant speed.

2. On a speed-time graph, what do these show?

- a). A horizontal line: the object is moving at a constant speed (or is stationary, if the line is along the time axis).
- b). A straight line sloping steeply upwards: the object is accelerating quickly (a large acceleration).
- c). A straight line sloping gently downwards: the object is decelerating slowly (a small deceleration).

Graphs of Motion – Section A (5 marks)

- 1. d) moving at a constant speed
- 2. c) its speed
- 3. b) accelerating
- 4. a) moving at a high speed
- 5. a) moving with a high acceleration

Graphs of Motion – Section B (6 marks)

1. Walker's distance-time graph (A→B→C→D)

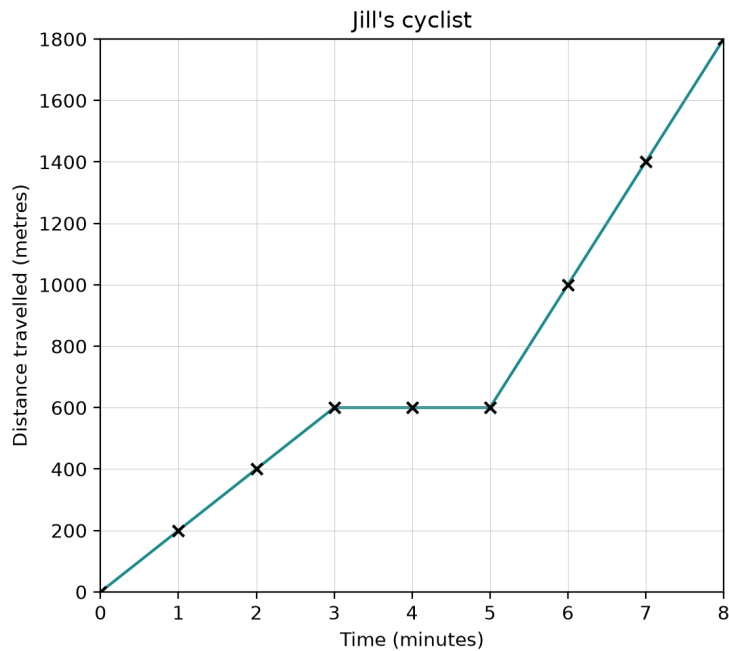
- a). B to C (the flat section).
- b). $250 - 125 = 125$ seconds
- c). A to B (the steepest section – 500 m in 125 s, faster than C to D's 500 m in 250 s).
- d). speed = $500 \text{ m} \div 125 \text{ s} = 4 \text{ m/s}$
- e). **1000 m** (the total distance reached by point D).
- f). average speed = $1000 \text{ m} \div 500 \text{ s} = 2 \text{ m/s}$

Graphs of Motion – Section C (20 marks)

1. Car's speed-time graph (A–F: 60 km/h, then stop, then 90 km/h)

- a). C to D (the section at zero speed).
- b). $1.0 - 0.5 = 0.5$ hours
- c). E to F (at 90 km/h, faster than the 60 km/h A–B section).
- d). **90 km/h**
- e). distance = $(60 \text{ km/h} \times 0.5 \text{ h}) + (0 \text{ km}) + (90 \text{ km/h} \times 1.0 \text{ h}) = 30 + 90 = 120 \text{ km}$
- f). average speed = $120 \text{ km} \div 2.0 \text{ h} = 60 \text{ km/h}$

2. Jill's cyclist (distance-time table, 0–8 min)



- b). From minute 3 to minute 5, the distance stays at 600 m, so the cyclist was stationary for **2 minutes**.
- c). The distance increases by 400 m in each of the intervals 5–6, 6–7 and 7–8 minutes – these three are tied for fastest.
- d). Greatest speed: 400 m in 1 minute (60 s) = $400 \div 60 \approx \mathbf{6.67 \text{ m/s}}$
- e). average speed = $1800 \text{ m} \div (8 \times 60 \text{ s}) = 1800/480 = \mathbf{3.75 \text{ m/s}}$

Pressure – Quick Test (p.97)

1. By applying a force over a small area (concentrating the force).
2. By spreading the force over a larger area.
3. The weight of the shopping is concentrated onto the small area of the thin handles, creating a high pressure on your hands.
4. Pascals (Pa).
5. A sharp knife has a much smaller (thinner) blade edge area, so the same force creates a much higher pressure, making it easier to cut through.
6. $P = F/A = 50 \div 2.5 = \mathbf{20 \text{ Pa}}$
7. Area = $4 \times 2 = 8 \text{ m}^2$. $P = 4000 \div 8 = \mathbf{500 \text{ Pa}}$
8. $F = P \times A = 40 \times 2 = \mathbf{80 \text{ N}}$
9. At the bottom of the lake (the greatest depth) – pressure increases with depth.

Pressure – Section A (5 marks)

1. c) someone wearing snowshoes
2. c) 16 Pa
3. b) 50 N
4. b) 5 m^2
5. b) increases with depth

Pressure – Section B (13 marks)

1. True or false (7 marks)

- a) Thin handles create high pressure, making it easier to carry – **False** (high pressure makes it more painful, not easier)
- b) Snowshoes spread force to create low pressure, stopping you sinking – **True**
- c) Pressure on a nail is greatest on the head – **False** (it's greatest at the sharp point, the smallest area)
- d) Sharp knives create higher pressures than blunt knives – **True**
- e) Press-ups on palms create the same pressure as on fingertips – **False** (fingertips have a much smaller area, so create much higher pressure for the same weight)
- f) Camels have large feet to grip the sand better – **False** (large feet spread their weight to reduce pressure, stopping them sinking – not about grip)
- g) More likely to fall through a frozen lake standing up than lying down – **True** (standing concentrates your weight on your feet; lying down spreads it over a larger area)

2. Calculate the pressure

- a). 50 N over 25 m² = $50/25 = 2$ Pa
- b). 80 N over 16 m² = $80/16 = 5$ Pa
- c). 25 N over 2.5 m² = $25/2.5 = 10$ Pa

3. Calculate the force

- a). over 10 m² for 3 Pa: $F = 3 \times 10 = 30$ N
- b). over 4 m² for 5 Pa: $F = 5 \times 4 = 20$ N
- c). over 16 m² for 1.5 Pa: $F = 1.5 \times 16 = 24$ N

Pressure – Section C (12 marks)

1. 600 N crate, faces A, B, C (3 m × 2 m × 1 m)

The exact face each letter refers to is hard to confirm precisely from this scan – this reading assumes A = the 2 m × 1 m side face (2 m²), B = the 3 m × 1 m side face (3 m²), and C = the 3 m × 2 m base (6 m²); check against your own diagram.

- a). Face A (2 m²): $P = 600/2 = 300$ Pa
- b). Face B (3 m²): $P = 600/3 = 200$ Pa
- c). Face C (6 m²): $P = 600/6 = 100$ Pa
- d). Face C should be in contact with the lawn – it has the largest area, so it creates the lowest pressure (least likely to damage the grass).

2. Drawing pin diagram

- a). Mark H at the sharp point of the pin (smallest area, force concentrated = high pressure).
- b). Mark L at the flat, broad head of the pin (largest area, force spread out = low pressure).

Note: This is a Collins "KS3 Science Success Guide" page-set ("Graphs of motion" and "Pressure"), reproduced here by Aristeia Academy Tuition, rather than an exam-board past paper – there is no publicly downloadable official markscheme for it. These answers were worked out directly from the notes and data given earlier in the booklet. The Pressure section is identical to Year 8 Booklet 8, so the same answers apply there too.